Protostar Climate Forecasting: Data Quality, Missing Data and Final Pipeline

1. Executive summary

This report covers the work after Report 2: a second round of forecasting experiments (Experiments 4 and 4b), a full audit of data correctness and missing data (Experiments 5 and 5b), and the final pipeline that combines them. Every performance figure is a historical backtest on pre-2019 data. None is a forecast of 2019–2025.

The data is sound. Our missing-day counts match the dataset authors' own `Miss` columns on all 28,052 city-years, row by row, and the four quality-valid counts (24,583 / 24,858 / 24,868 / 24,235) reproduce exactly. The daily records contain no physically impossible temperatures. There are 9 rows with Tmax below Tmin, only one of them since 1989.

Most flagged anomalies are real weather. We checked for stuck sensors, rain recorded as zero and annual spikes; the flags turned out to be mostly genuine extremes or very stable maritime climates. Excluding flagged years never improved forecasts for any target, so flagged years are kept and documented rather than removed.

Two real issues were found, and their effect was measured.

- *Station changes:* 153 level shifts relative to neighbouring cities since 1950, 85% of them undocumented in `city_info`.
- *Missing summer days* in years that pass the 10-missing-day rule. This affects mainly older years: the likely CDD undercount averages only 0.7 degree-days per year in 2009–2018.

Data handling matters little, except for the temperature-based targets. Across eight strategies tested on identical truth values, only cooling degree days (CDD) improved meaningfully: by 1.0% with climatological infill plus homogenisation. Avg Temp improved by 0.4% with the same treatment. Precipitation and hottest day are best left under the strict rule.

The largest forecasting error was a modelling issue, not a data issue. Recent-mean forecasts lag a warming climate by about 8.5 years, which produced a systematic cold bias (CDD −116 degree-days on 2012–2018). Lag-corrected and delta-method forecasters remove most of that bias. Some cold bias remains in the most recent years, because warming ran faster than its own 30-year trend.

Final policy:

Target	Training labels	Forecaster family
Avg Temp	≤30 missing days with infill, homogenised	Lag-corrected recent mean
CDD	≤30 missing days with infill, homogenised	Lag-corrected mean or delta method (decided by the final backtest)
Max Tmax	Strict ≤10 missing days	30-year median or recent mean
Precipitation	Strict ≤10 missing days	30-year median or full-record mean

2. Data and scope

The challenge dataset (`CMU-OSPO/protostar-climate-train-to-2018`; Lai & Dzombak, 2019) provides daily Tmax, Tmin and precipitation for 210 U.S. cities from 1851 to 2018 (10,245,754 daily rows), annual indices for 28,052 city-years, and station-history metadata (`city_info`, 461 station periods). The four forecast targets are annual mean temperature (Avg Temp), total precipitation, hottest-day temperature (Max Tmax) and cooling degree days (CDD, base 65 °F).

The following rules applied throughout:

- No data dated after 2018-12-31 is loaded. Every notebook asserts this.
- All statistics, climatologies, breaks and slopes are computed from data up to each backtest origin.
- Truth values are always the strict official annual values, identical across all compared methods and strategies.
- Each notebook includes a unit test showing that scrambling every value after an origin changes no prediction.

3. Data correctness checks

3.1 Reconciliation

The strict rule (at most 10 missing days, where "missing" means an absent date or a NaN value) reproduces the four reconciled counts exactly. More importantly, our per-year missing-day counts equal the dataset's `Miss Tmax`, `Miss Tmin` and `Miss Prcp` columns on **28,052 of 28,052** rows. That confirms the counting rule is identical to the dataset authors' and settles the discrepancy first raised in Report 2.

3.2 Daily integrity

Check	Rows	Since 1989
Tmax < Tmin	9	1
Tmax > 130 °F or Tmin < −70 °F	0	0
Precipitation < 0	0	0
Precipitation > 20 in/day	1	0
Tmax = Tmin, both present	550	8

The handful of inconsistencies are old and too few to affect annual values.

3.3 Stuck-sensor runs

A first test flagged 211 runs of seven or more identical daily Tmax values in 32 cities. They were concentrated in Galveston, Key West, Miami, West Palm Beach and Corpus Christi, mostly in July, where summer highs genuinely hold at 89–91 °F for days. A revised test judges each run against the city's own day-to-day persistence in that month. It still flags 43 runs, 8 of them since 1989, and the recent ones are borderline (the strongest, Galveston in July 2016, has an expected count of 0.018). Because the test assumes independent days, it understates real persistence. The remaining recent flags therefore probably reflect weather too.

3.4 Precipitation recorded as zero

The 43 precipitation years that differ sharply from their neighbours (30 too dry, 13 too wet) were examined. Only 2 show the zero-coding signature of a dry year accompanied by many extra zero-rain days: USW00024018 (1876) and USW00094129 (2005). The station with the most extreme dry years, USW00003145, is a desert site with 349 dry days in a typical year. Its record lows are consistent with real drought, so they are not treated as errors.

3.5 Annual spikes

273 annual temperature or CDD values deviate from their neighbours by more than five robust standard deviations, measured against a local nine-year median so that station shifts are not mistaken for spikes. Removing them made Avg Temp, Max Tmax and CDD forecasts slightly worse. They are kept as genuine extremes.

4. Missing data: which days, and how much bias

A year can pass the 10-day rule while missing ten July days. Summing the remaining days then undercounts CDD and precipitation, and can miss the hottest day. For every missing day we estimated the likely value from the city's mean for that calendar day over the previous 10 years (smoothed over ±7 days; no future data).

Effect on strict-valid years with at least one missing day	Mean	95th percentile	Maximum
CDD likely missed (degree-days)	8.3	37.7	168
Precipitation likely missed (inches)	0.24	0.94	3.3
Avg Temp shift (°F)	−0.05	±0.36	1.19

1,477 of the 4,417 strict-valid CDD years with missing days lack at least one summer day, and in 129 city-years the missing days probably held more than 50 degree-days. For hottest day, 533 years are missing at least one day in the city's 30 climatologically hottest days.

The bias sits mainly in older records, however. In 2009–2018, the likely CDD undercount averages **0.7 degree-days** per strict-valid year, and only 0.9% of those years exceed 25. Recent levels, which the forecasts depend on most, are almost unaffected. The exceptions are two data-poor cities: USC00380072 (31 degree-days) and USC00280734 (17).

5. Station inconsistencies

Each city's annual anomaly was compared with the median anomaly of its eight nearest neighbours, and steps in that difference were detected ($t \geq 5$; at least 0.6 °F or 120 degree-days; segments of at least five years). Since 1950 this found 165 shifts, 12 of which are echoes: one city's real break appears with the opposite sign in every city that uses it as a neighbour. An earlier audit showed this clearly in precipitation, where three Pacific Northwest cities echoed a single 27% drop at USW00024221 in 1996–98.

Target	Real shifts	Since 1999	Match a documented transition
Avg Temp	86	17	14

Target	Real shifts	Since 1999	Match a documented transition
CDD	52	16	7
Precipitation	8	2	1
Max Tmax	7	1	1

Only 15% match a transition in `city_info`. That metadata records changes of station ID, not relocations or instrument changes under the same ID, so the low share does not mean the rest are false. Twenty-five cities have an Avg Temp or CDD shift since 1999, inside the 30-year window used for trend estimation. The largest is USW00023185, +4.1 °F in 1999, consistent with rapid urban growth around the station. USW00093134 shows a CDD drop of 461 degree-days in 1999, matching the widely reported relocation of the downtown Los Angeles station that year.

Forecasts need the level of the *current* station. Homogenisation therefore shifts the years before each break to the post-break level instead of deleting them. An early test on backtest values showed that this helps CDD but harms precipitation and hottest day, where most detections are noise. That is why the final policy applies homogenisation only to Avg Temp and CDD.

6. Strategy test

Each strategy changes only the history the forecasters learn from. Truth values are the strict official values on matched city-years, and the evaluation uses annual origins 1990–2011 with seven-year horizons. P(better) comes from a bootstrap that resamples calendar years, because errors are strongly correlated across cities within a year.

Strategy	Avg Temp ΔMAE	Precip ΔMAE	Max Tmax ΔMAE	CDD ΔMAE
S0: strict ≤10 days (reference MAE)	1.159 °F	5.797 in	2.660 °F	165.88
S1: ≤5 days	+0.006	−0.005 (P 0.99)	+0.009	+0.17
S2: ≤20 days, raw	−0.002	+0.002	−0.001	−0.32 (P 0.96)
S3: ≤30 days, infilled	−0.001	+0.003	−0.003 (P 0.99)	−0.67 (P 1.00)
S4: strict + infill	+0.000	+0.001	0	−0.13
S5: strict + homogenised	−0.003 (P 0.96)	0	0	−0.86 (P 1.00)
S6: strict − flagged years	+0.000	0.000	+0.001	+0.12
S7: ≤30 infilled + homogenised	**−0.005 (P 0.997)**	+0.003	−0.003	**−1.59 (P 1.00)**

The CDD gains from infill and homogenisation are separate and add up (−0.67 and −0.86, then −1.59 combined), which fits the diagnosis: infill restores missing summer days, and homogenisation removes station steps. For precipitation and hottest day, the differences are thousandths of an inch or degree. With about 31,000 forecasts they are statistically detectable, but they are practically irrelevant. A measured "no change helped" is itself a result worth stating in the data card.

The remaining CDD cold bias (about −30 degree-days) is identical under every strategy. It comes from the forecasting model, not from the data.

7. Forecasting methods and backtest history

Experiments 1–3 (Report 2) used four non-overlapping origins. Recent 10-year means were the strongest simple baseline, but the experiments never reported signed bias.

Experiment 4 used annual rolling origins from 1975 to 2011 and showed the bias clearly. A recent 10-year mean is centred about 4.5 years before the origin, and forecasts are on average 4 years after it, so it lags the climate by about 8.5 years. Its bias over origins 1990–2011 was −0.33 °F for Avg Temp and −53 degree-days for CDD, rising to about −0.45 °F and −80 to −105 degree-days in origins since 2000. Two new methods corrected most of this:

- the **lag-corrected mean**: the recent level plus a neighbour-pooled 30-year slope multiplied by the lag;
- the **delta method** for CDD: recent daily temperatures shifted by the projected warming, with CDD then recomputed.

Origins 1990–2011	MAE	Bias
Avg Temp, recent 10-year mean	1.172	−0.33
Avg Temp, lag-corrected (a = 0.75)	1.160	−0.04
Avg Temp, 30-year trend	1.191	+0.02
CDD, recent 10-year mean	165.7	−53
CDD, delta method (a = 1)	164.8	+5
CDD, 30-year trend	170.2	−24

The one-standard-error selection rule picked the simplest method by construction, because the bootstrap standard error was as large as the entire spread between candidates. Experiment 4b replaced that choice with an explicit override, justified by era tables: recent means win only in the slow-warming 1975–1999 origins, and lag-corrected and delta methods win in every period since 2000. For precipitation and hottest day, a 30-year median beat the recent mean with probability 0.99 and 0.95.